Yifan Chen

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Employment

Department of Mathematics, UCLA — Los Angeles, CA
Assistant Professor (tenure track) — 2025/7–present

DataX, UCLA — Los Angeles, CA
Affiliated Faculty in Fundamental Data Science — 2025/7–present

Courant Institute, New York University — New York, NY
Assistant Professor/Courant Instructor (non-tenure track) — 2023/9–2025/7
Hosted by Eric Vanden-Eijnden and Benjamin Peherstorfer

Education

California Institute of Technology — Pasadena, CA
Ph.D. in Applied and Computational Mathematics — 2018–2023
Advisors: Thomas Y. Hou, Houman Owhadi, Andrew M. Stuart

Tsinghua University — Beijing, China
B.S. in Pure and Applied Mathematics — 2014–2018

Research Interests

My research lies at the interface of applied and computational mathematics, applied probability, and statistics. I study the interplay between probability, dynamics, and numerics for high-dimensional and large-scale scientific computing, data science, and machine learning. My recent work focuses on high dimensional stochastic methods for sampling and generative modeling, and on probabilistic perspectives and algorithms for PDEs and linear algebra.

Publications

Journal publications

[1] **Yifan Chen**, Eric Vanden-Eijnden. Scale-Adaptive Generative Flows for Multiscale Scientific Data. *Accepted to SIAM Journal on Scientific Computing*, 2026.

[2] José A. Carrillo, **Yifan Chen**, Daniel Zhengyu Huang, Jiaoyang Huang, Dongyi Wei. Fisher-Rao Gradient Flow: Geodesic Convexity and Functional Inequalities. *SIAM Journal on Mathematical Analysis*, 2026.

[3] **Yifan Chen**, Xiaoou Cheng, Jonathan Niles-Weed, Jonathan Weare. Convergence of Unadjusted Langevin in High Dimensions: Delocalization of Bias. *Communications on Pure and Applied Mathematics*, 2026.

[4] **Yifan Chen**, Daniel Zhengyu Huang, Jiaoyang Huang, Sebastian Reich, Andrew M. Stuart. Sampling via Gradient Flows in the Space of Probability Measures. *Mathematics of Computation*, 2026.

[5] Huan Zhang, **Yifan Chen**, Eric Vanden-Eijnden, Benjamin Peherstorfer. Sequential-in-time training of nonlinear parametrizations for solving time-dependent partial differential equations. *SIAM Review Research Spotlights*, 2026.

[6] Baojun Che, **Yifan Chen**, Zhenghao Huan, Daniel Zhengyu Huang, Weijie Wang. Stable Derivative Free Gaussian Mixture Variational Inference for Bayesian Inverse Problems. *SIAM Journal on Scientific Computing*, 2025.

[7] **Yifan Chen**, Ethan N. Epperly, Joel A. Tropp, and Robert J. Webber. Randomly pivoted Cholesky: Practical approximation of a kernel matrix with few entry evaluations. *Communications on Pure and Applied Mathematics*, 2024.

[8] **Yifan Chen**, Bamdad Hosseini, Houman Owhadi, Andrew M. Stuart. Gaussian Measures Conditioned on Nonlinear Observations: Consistency, MAP Estimators, and Simulation. *Statistics and Computing*, 2024.

[9] **Yifan Chen**, Daniel Zhengyu Huang, Jiaoyang Huang, Sebastian Reich, Andrew M. Stuart. Efficient, Multimodal, and Derivative-Free Bayesian Inference With Fisher-Rao Gradient Flows. *Inverse Problems*, 2024.

[10] Pau Batlle, **Yifan Chen**, Bamdad Hosseini, Houman Owhadi, Andrew M. Stuart. Error Analysis of Kernel/GP Methods for Nonlinear and Parametric PDEs. *Journal of Computational Physics*, 2024.

[11] Yu Sun, Zihui Wu, **Yifan Chen**, Berthy T. Feng, Katherine L. Bouman. Provable Probabilistic Imaging using Score-Based Generative Priors. *IEEE Transactions on Computational Imaging*, 2024.

[12] **Yifan Chen**, Houman Owhadi, Florian Schaefer. Sparse Cholesky Factorization for Solving Nonlinear PDEs via Gaussian Processes. *Mathematics of Computation*, 2024.

[13] **Yifan Chen**, Thomas Y. Hou, and Yixuan Wang. Exponentially convergent multiscale methods for 2D high frequency heterogeneous Helmholtz equations. *SIAM Multiscale Modeling & Simulation*, 21(3): 849–883, 2023.

[14] **Yifan Chen**, Thomas Y. Hou, and Yixuan Wang. Exponentially convergent multiscale finite element method. *Communications on Applied Mathematics and Computation*, 1–17, 2023.

[15] **Yifan Chen**, Thomas Y. Hou. Multiscale elliptic PDE upscaling and function approximation via subsampled data. *SIAM Multiscale Modeling & Simulation*, 20(1):188–219, 2022.

[16] **Yifan Chen**, Bamdad Hosseini, Houman Owhadi, and Andrew M. Stuart. Solving and learning nonlinear PDEs with Gaussian processes. *Journal of Computational Physics*, 447:110668, 2021.

[17] **Yifan Chen**, Houman Owhadi, and Andrew M. Stuart. Consistency of empirical Bayes and kernel flow for hierarchical parameter estimation. *Mathematics of Computation*, 90(332):2527–2578, 2021.

[18] **Yifan Chen**, Thomas Y. Hou, and Yixuan Wang. Exponential convergence for multiscale linear elliptic PDEs via adaptive edge basis functions. *SIAM Multiscale Modeling & Simulation*, 19(2):980–1010, 2021.

[19] **Yifan Chen**, Thomas Y. Hou. Function approximation via the subsampled Poincaré inequality. *Discrete & Continuous Dynamical Systems-A*, 41(1), 2021.

[20] **Yifan Chen**, Wuchen Li. Optimal transport natural gradient for statistical manifolds with continuous sample space. *Information Geometry*, 3(1):1–32, 2020.

[21] **Yifan Chen**, Yuejiao Sun, and Wotao Yin. Run-and-Inspect Method for nonconvex optimization and global optimality bounds for R-local minimizers. *Mathematical Programming*, 176(1): 39-67, 2019.

[22] Jing Chen, **Yifan Chen**, Hao Wu, and Dinghui Yang. The quadratic Wasserstein metric for earthquake location. *Journal of Computational Physics*, 373:188–209, 2018.

Conference publications

[1] Jiawei Xu, Minghui Liu, Aakriti Agrawal, **Yifan Chen**, Furong Huang. Scheduling Thoughts: Learning the Order of Thought in Diffusion Language Models. *ICML*, 2026.

[2] Wenda Chu, Zihui Wu, **Yifan Chen**, Yang Song, Yisong Yue. Split Gibbs Discrete Diffusion Posterior Sampling. *NeurIPS*, 2025.

[3] **Yifan Chen**, Mark Goldstein, Mengjian Hua, Michael S. Albergo, Nicholas M. Boffi, Eric Vanden-Eijnden. Probabilistic Forecasting with Stochastic Interpolants and Föllmer Processes. *ICML*, 2024.

[4] Xinzhe Dai, Peichen Zhong, Bowen Deng, **Yifan Chen**, and Gerbrand Ceder. Inpainting crystal structure generations with score-based denoising. *ICML Workshop AI4Science*, 2024.

[5] Zihui Wu, Yu Sun, **Yifan Chen**, Bingliang Zhang, Yisong Yue, Katherine L. Bouman. Principled Probabilistic Imaging using Diffusion Models as Plug-and-Play Priors. *NeurIPS*, 2024.

Preprints

[1] Hong Ye Tan, **Yifan Chen**. Geometric Ergodicity of Affine Invariant Ensemble Langevin and its Discrete Time Variants, *arXiv: 2609.03326*, 2026.

[2] Evan Bell, Jiaming Liu, **Yifan Chen**, Yu Sun. Generative Translation Priors: Bayesian Imaging with Cross-Modality Image Translation, *arXiv: 2608.28872*, 2026.

[3] Deliang Wei, Evan Bell, Wenhan Guo, **Yifan Chen**, Yu Sun. Picard Proximal Monte Carlo for Parallel Bayesian Imaging with Score-Based Generative Priors, *arXiv: 2608.17666*, 2026.

[4] **Yifan Chen**, Xiaoou Cheng, Jonathan Niles-Weed, Jonathan Weare. Delocalization of Bias in Unadjusted Hamiltonian Monte Carlo and Underdamped Langevin. *arXiv: 2607.15208*, 2026.

[5] **Yifan Chen**, Eric Vanden-Eijnden. Variational Optimality of Föllmer Processes in Generative Diffusions. *arXiv: 2602.10989*, 2026.

[6] Baojun Che, **Yifan Chen**, Daniel Zhengyu Huang, Xinying Mao, Weijie Wang. Adaptive Exponential Integration for Stable Gaussian Mixture Black-Box Variational Inference. *arXiv: 2601.14855*, 2026.

[7] **Yifan Chen**, Sifan Liu. Rotated Mean-Field Variational Inference and Iterative Gaussianization. *arXiv: 2510.07732*, 2025.

[8] **Yifan Chen**, Eric Vanden-Eijnden, Jiawei Xu. Lipschitz-Guided Design of Interpolation Schedules in Generative Models. *arXiv: 2509.01629*, 2025.

[9] **Yifan Chen**. New Affine Invariant Ensemble Samplers and Their Dimensional Scaling. *arXiv: 2505.02987*, 2025.

Grants

- NSF DMS-2608264: Mathematical Foundations and Scalable Algorithms for High-Dimensional Sampling and Generative Modeling (PI, $300K) 2026–2029
- DataX Pilot Projects Grant Program, UCLA DataX (Lead PI, with co-PI Erik Petigura in astrophysics, $34K) 2026
- Research Allowance Program, UCLA Council on Research Faculty Grants ($5K) 2026
- AMS-Simons Travel Grant, American Mathematical Society and Simons Foundation ($5K) 2025–2027

Awards

- The W.P. Carey and Co. Prize in Applied Mathematics, Caltech 2023
- Kortschak Scholars Program, Department of Computing and Mathematical Sciences, Caltech 2018–2023
- Tsinghua Xuetang Mathematics Program, Department of Mathematical Sciences, Tsinghua University 2015–2018
- Outstanding Undergraduate, Tsinghua University and Beijing 2018
- Baosteel Scholarship, Baosteel Corporation 2017
- Scholarship in Memory of the "12.9" Student Movement, Tsinghua University 2016
- Qualcomm Scholarship, Qualcomm Corporation 2016
- Scholarship in Memory of Mathematics Professor Ou Li, Tsinghua University 2016
- National Scholarship, Ministry of Education of China 2015

Teaching

Instructor at UCLA

- MATH 285J: Seminar in Applied Mathematics (Topics in High-Dimensional Sampling and Generative Diffusions) Spring 2026
- MATH 164: Optimization Winter 2026

Instructor at NYU Courant

- MATH-UA 252: Numerical Analysis Spring 2025
- MATH-UA 252: Numerical Analysis Fall 2024
- MATH-UA 120: Discrete Mathematics Spring 2024
- MATH-UA 120: Discrete Mathematics Fall 2023

Teaching Assistant at Caltech

- ACM 109: Mathematical Modeling Spring 2021
- ACM 118: Stochastic Processes and Regression Winter 2021
- ACM 117: Probability and Stochastic Processes Fall 2020
- ACM 109: Mathematical Modeling Spring 2020

Industrial Experiences

Citadel Securities Miami, FL
Quantitative Research Intern 2022/6–2022/8

- Project on Alpha Research: Predicting APAC Market Returns

Microsoft (virtual) Redmond, WA
Part Time Researcher, working with Pengchuan Zhang 2021/9–2022/2

- Project: Stabilizing Large Scale Neural Network Training of Vision Transformers

Referee Services

- **Journals**: Analysis and Applications, Annals of Applied Probability, Communications on Pure and Applied Mathematics, Computational Methods in Applied Mathematics, Computers & Fluids, Discrete and Continuous Dynamical Systems, European Journal of Applied Mathematics, Foundations of Data Science, IMA Journal of Numerical Analysis, International Journal of Computer Mathematics, Journal of Computational Physics, Journal of Functional Analysis, Journal of Machine Learning Research, Journal of the Royal Statistical Society Series B, Linear Algebra and Its Applications, Mathematics of Computation, Nature Machine Intelligence, Proceedings of the National Academy of Sciences (PNAS), Research in the Mathematical Sciences, SIAM Journal on Control and Optimization, SIAM Journal on Imaging Sciences, SIAM Journal on Mathematics of Data Science, SIAM Journal on Numerical Analysis, SIAM Journal on Optimization, SIAM Multiscale Modeling & Simulation, SIAM/ASA Journal on Uncertainty Quantification
- **Conferences**: NeurIPS (outstanding reviewer), International Conference on Geometric Science of Information (GSI)

Service and Organization of Sessions, Workshops, and Conferences

- Advanced to Candidacy committee: Yifan Gu (2026; advisor: Andrea Bertozzi)
- Qualifying exam committee: numerical analysis
- Organizing committee, UCLA Math Colloquium 2025-2026

- Organizing committee, UCLA Data Theory Seminar 2025-
- Graduate admission committee, UCLA Math 2025-2026
- Co-organizer, minisymposium "Analysis of Langevin and Related Algorithms", International Conference on Monte Carlo Methods and Applications, Illinois Institute of Technology Chicago, IL · Jul 2025
- Co-organizer, minisymposium "Efficient Computation and Learning with Randomized Sampling and Pruning", SIAM Conference on Mathematics of Data Science (MDS24) Atlanta, GA · Oct 2024
- Co-organizer, minisymposium "Recent Advances in Kernel Methods for Computing and Learning", SIAM Conference on Mathematics of Data Science (MDS22) San Diego, CA · Sep 2022

Conferences and Seminars

- "Generative diffusion and sampling: algorithms and theory in high dimensions", DataX+IDRE AI Symposium, UCLA Los Angeles, CA · May 2026
- "Affine Invariant Samplers and Flows: Analysis and New Algorithms", Monte Carlo Online Seminar Virtual · Apr 2026
- "Affine Invariant Samplers and Flows: Analysis and New Algorithms", SIAM Conference on Uncertainty Quantification (UQ26) Minneapolis, MN · Mar 2026
- "Variational Optimality of Föllmer Processes and related design questions in generative diffusions", Generative Modeling Reading Group Seminar, Duke University Virtual · Mar 2026
- "Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!", Mathematics of Information, Data, and Signals Seminar, UCSD San Diego, CA · Feb 2026
- NeurIPS 2025 San Diego, CA · Dec 2025
- "Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!", Applied Mathematics Seminar, University of California, Berkeley Berkeley, CA · Nov 2025
- "Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!", Applied Mathematics Colloquium, UCLA Los Angeles, CA · Oct 2025
- "Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!", Scientific Computing and Numerics Seminar, Cornell University Virtual · Oct 2025
- "Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!", Computational Math + X Seminar, Caltech Pasadena, CA · Oct 2025
- "Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!", PDE & Applied Mathematics Seminar, UC Riverside Virtual · Oct 2025
- "Exploring High Dimensions in Dynamical Sampling: Flattening the Scaling Curve!", Computational and Applied Mathematics Colloquium, Penn State University University Park, PA · Sep 2025
- "Lipschitz Guided Dynamical Transport for generative flows of ill-conditioned distributions", Workshop on Scientific Machine Learning: Theory, Algorithms, and Applications, Purdue University West Lafayette, IN · Sep 2025
- "Convergence of Unadjusted Langevin in High Dimensions: Delocalization of Bias", International Conference on Monte Carlo Methods and Applications, Illinois Institute of Technology Chicago, IL · Jul 2025
- "New Affine Invariant Ensemble Samplers and Their Dimensional Scaling", Computational Bayesian Statistics Journal Club, Flatiron Institute New York, NY · May 2025
- "Analysis and design of high dimensional sampling with applications in scientific computing", Purdue-PSU-Maryland Joint Mathematical Data Science Seminar Virtual · Apr 2025
- "Analysis and design of high dimensional sampling with applications in scientific computing", Applied Math Colloquium, UCLA Los Angeles, CA · Jan 2025
- "Convergence of Unadjusted Langevin in High Dimensions: Delocalization of Bias", Probability Seminar, University of Michigan Ann Arbor, MI · Jan 2025
- "Analysis and design of high dimensional sampling with applications in scientific computing", Colloquium, University of Michigan Ann Arbor, MI · Jan 2025

- "Analysis and design of high dimensional sampling with applications in scientific computing", Applied Math Colloquium, Duke University — Durham, NC · Jan 2025
- "Analysis and design of high dimensional sampling with applications in scientific computing", CMOR Special Lecture, Rice University — Houston, TX · Dec 2024
- "Randomized and sparse Cholesky for learning with Gaussian processes", SIAM Conference on Mathematics of Data Science (MDS24) — Atlanta, GA · Oct 2024
- Recent Advances and Future Directions for Sampling, Yale University — New Haven, CT · Oct 2024
- "On approximate, efficient sampling in high dimensional scientific computing", Data Science Seminar, University of Minnesota — Minneapolis, MN · Oct 2024
- "On approximate, efficient sampling in high dimensional scientific computing", Rising Stars in Computing for Science and Engineering, Harvard University — Cambridge, MA · Sep 2024
- "Convergence of Unadjusted Langevin in High Dimensions: Delocalization of Bias", Computational Bayesian Statistics Journal Club, Flatiron Institute — New York, NY · Sep 2024
- "Fast, Multimodal, Derivative-Free Bayes Inference with Fisher-Rao Gradient Flows", Applied Mathematics Seminar, Nanyang Technological University — Singapore · Jul 2024
- "Probabilistic Forecasting with Stochastic Interpolants and Föllmer Processes", International Conference on Scientific Computation and Differential Equations, National University of Singapore — Singapore · Jul 2024
- Conference on Multiscale Modeling based on Physics and Data, IPAM, UCLA — Los Angeles, CA · Apr 2024
- "Fast, Multimodal, Derivative-Free Bayes Inference with Fisher-Rao Gradient Flows", Applied Math Colloquium, Columbia University — New York, NY · Jan 2024
- "Fast, Multimodal, Derivative-Free Bayes Inference with Fisher-Rao Gradient Flows", Workshop on Scientific Computing and Large Data, University of South Carolina — Columbia, SC · Dec 2023
- "Fast, Multimodal, Derivative-Free Bayes Inference with Fisher-Rao Gradient Flows", Numerical Analysis Seminar, The University of Hong Kong — Virtual · Dec 2023
- Measure Transport, Diffusion Processes and Sampling Workshop, Flatiron Institute — New York, NY · Dec 2023
- "On Multiscale and Statistical Numerical Methods for PDEs and Inverse Problems", Courant Postdoc Day, Courant Institute — New York, NY · Nov 2023
- "Design of Gradient Flows for Sampling: Energy Functionals, Invariance, and Gaussian Approximation", Yau Mathematical Sciences Center CAM Seminar, Tsinghua University — Virtual · Nov 2023
- International Workshop on Recent Developments in Applied Mathematics and its Applications, Caltech — Pasadena, CA · Nov 2023
- "Design of Gradient Flows for Sampling: Energy Functionals, Invariance, and Gaussian Approximation", Scientific Machine Learning Seminar, Courant Institute — New York, NY · Oct 2023
- "Solving and Learning Nonlinear PDEs with Gaussian Processes", 17th U.S. National Congress on Computational Mechanics — Albuquerque, NM · Jul 2023
- "Sparse Cholesky Factorization for Solving Nonlinear PDEs via Gaussian Processes", Mathematical and Scientific Machine Learning, ICERM — Providence, RI · Jun 2023
- "Design of Gradient Flows for Sampling: Energy Functionals, Invariance, and Gaussian Approximation", AIMS Conference on Dynamical Systems, Differential Equations and Applications — Wilmington, NC · May 2023
- "Gradient flows for sampling: affine invariance and numerical approximation", Southern California Applied Mathematics Symposium, UC Irvine — Irvine, CA · Apr 2023
- "Developments of Multiscale and Probabilistic Methods for Solving PDEs and Inverse Problems", Applied Math Colloquium, Peking University — Virtual · Feb 2023
- "Developments of Multiscale and Probabilistic Methods for Solving PDEs and Inverse Problems", Applied Math Colloquium, Columbia University — Virtual · Jan 2023
- International Conference on New Trends in Computational and Data Sciences, Caltech — Pasadena, CA · Dec 2022
- "Solving and Learning Nonlinear PDEs with Gaussian Processes", SIAM Conference on Mathematics of Data Science (MDS22) — San Diego, CA · Sep 2022

- "Solving and Learning Nonlinear PDEs with Gaussian Processes", Southern California Applied Mathematics Symposium, Harvey Mudd College — Claremont, CA · May 2022
- "Solving and Learning Nonlinear PDEs with Gaussian Processes", Rough Path Interest Group, The Alan Turing Institute — Virtual · Apr 2022
- "Solving and Learning Nonlinear PDEs with Gaussian Processes", minisymposium "New Developments in Gaussian Processes", SIAM Conference on Uncertainty Quantification (UQ22) — Atlanta, GA · Apr 2022
- "Solving and Learning Nonlinear PDEs with Gaussian Processes", CMX Student and Postdoc Seminar, Caltech — Pasadena, CA · Nov 2020
- "Consistency of Hierarchical Parameter Learning: Empirical Bayesian and Kernel Flow Approaches", Second Symposium on Machine Learning and Dynamical Systems, Fields Institute — Virtual · Sep 2020
- "Consistency of Hierarchical Parameter Learning: Empirical Bayesian and Kernel Flow Approaches", Bernoulli-IMS One World Symposium — Virtual · Aug 2020
- Oberwolfach Seminar: Beyond Numerical Homogenization — Oberwolfach, Germany · Jun 2019
- Machine Learning for Multiscale Model Reduction Workshop, Harvard University — Cambridge, MA · Mar 2019
- Mathematical Model and Computation of Nonlinear Problems, Tsinghua Sanya International Mathematics Forum — Sanya, China · Jan 2018
- "The Quadratic Wasserstein Metric for Earthquake Location", Youth Forum, 15th Annual Meeting of CSIAM — Qingdao, China · Oct 2017

Computer Skills

Professional experiences in Python, Julia, MATLAB.

Languages

English (fluent), Chinese (native)